

Object Oriented Programming Lab

Lab 10

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

You are required to design an employee salary evaluation system using **inheritance** and **polymorphism**. Create base class **Employee** with following attributes and functions:

- Data members (protected):
 - string name
 - int id
- Member functions:
 - A constructor to initialize data members
 - virtual double getSalary() {}
 - A function void showInfo() to display employee name and id

Create Three Derived Classes. Each class must inherit publicly from **Employee** and override **getSalary()**.

Teacher

- Additional attribute: double payPerHour, int hours
- getSalary() returns: payPerHour * hours

Manager

- Additional attribute: double fixedSalary, double bonus
- getSalary() returns: fixedSalary + bonus

Staff

- Additional attribute: double monthlySalary
- getSalary() returns: monthlySalary

In main()

1. Declare a base class reference Employee* emp;
2. Create objects of child classes, assign each object to the base-class reference and make a call to getSalary().
Observe that the correct version of getSalary() executes for each object.

Expected Output (Example)

Employee: Ali (ID: 101)

Salary: 45000

Employee: Sara (ID: 201)

Salary: 85000

Employee: Hamza (ID: 301)

Salary: 30000